



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)

EXAMINATIONS - JAN/FEB 2023

SUBJECT : MAT 2152 ENGG. MATHS III
(MAKEUP)

ENGINEERING MATHEMATICS III [MAT 2152]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Expand $f(x) = x^2, 0 < x < 2, f(x+2) = f(x)$ as a Fourier Series and hence show that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ at $x=0$. (4)
- 1B) Expand $f(x) = 2x - 1, 0 < x < 1$ as a half-range Fourier sine series. (3)
- 1C) Find the Fourier transform of $f(x) = \begin{cases} 1, & |x| \leq a \\ 0, & |x| > a \end{cases}$ and hence evaluate $\int_0^{\infty} \left(\frac{\sin t}{t}\right)^2 dt$. (3)
- 2A) Find the analytic function $f(z) = u + iv$, where
$$u = \frac{y}{x^2 + y^2}$$
 (4)
- 2B) Show that $f(z) = e^{-z}$ is analytic. Find $f'(z)$. (3)
- 2C) Find the Fourier sine transform of the following function,
$$f(x) = e^{(-ax)}, a > 0$$
 (3)
- 3A) Evaluate $\oint_C \frac{z^2+5}{(z-2)(z-3)} dz$ using residue theorem, C is the circle $|z| = 4$ (4)
- 3B) Evaluate the following integral, $\int_C \frac{z^2-z+1}{z-2} dz$ Where C is
(i) $|z-1| = 1/2$ (ii) $|z| = 2$ (iii) $|z| = 1/2$ (3)
- 3C) Find Laurent Series expansion of $\frac{z^2-1}{z^2+5z+6}$, about (i) $|z| < 2$ (ii) $2 < |z| < 3$ (3)
- 4A) (4)

Verify Green's theorem for

$$\oint_C (xy + y^2) dx + x^2 dy,$$

where C is the boundary of the region defined by $y = x$, $y = x^2$

4B) Find the equation of the tangent plane and normal line to the surface $x^2 + y^2 = z$ at $(2, -1, 5)$. (3)

4C) (a) Show that $\vec{F} = (2xy + z^2)\hat{i} + (x^2)\hat{j} + (3xz^2)\hat{k}$ is a conservative force field.
(b) Find its scalar potential. (3)
(c) Find the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4)$.

5A) Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} - 2 \frac{\partial^2 u}{\partial y^2} = 0$ using the indicated transformations (4)
 $v = x + y, z = 2x - y.$

5B) Assuming the most general solution, Solve the heat equation $u_t = c^2 u_{xx}, 0 \leq x \leq \pi$ to find the temperature distribution at any time, subjected to the condition (3)
 $u(0, t) = 0, u(\pi, t) = 0$ and
 $u(x, 0) = \begin{cases} x & \text{in } 0 \leq x \leq \frac{\pi}{2} \\ \pi - x & \text{in } \frac{\pi}{2} \leq x \leq \pi \end{cases}$

5C) Use Divergence theorem to evaluate $\iint_S \vec{F} \cdot \vec{n} \, ds$. Where $\vec{F} = 4xzi - y^2j + yzk$ and S is the surface of the cube bounded by $x = 0, x = 2, y = 0, y = 2, z = 0, z = 2$. (3)

-----End-----